ModelIt!

Teacher's Guide

Restoring Ecosystems — Data-Driven Recovery

Using Science to Heal Damaged Landscapes

Quick Reference

Grade Level	Grade NE
NGSS Standard	HS-LS2-6, HS-LS2-7, HS-ESS3-3, HS-ESS3-4, HS-ETS1-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-6, HS-LS2-7, HS-ESS3-3, HS-ESS3-4, HS-ETS1-4

Evaluate claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent numbers and types of organisms in stable conditions, but changing conditions may result in a new ecosystem. Design, evaluate, and refine a solution for reducing the impacts of human activities on the environment and biodiversity. Create a computational simulation of a natural phenomenon. Evaluate or refine a technological solution that reduces impacts of human activities on natural systems. Use a computer simulation to model the impact of proposed solutions to a complex real-world problem.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a 9-component computational model of ecosystem restoration with reinforcing feedback loops, tipping points, and competing pressures to evaluate restoration strategies.
Disciplinary Core Idea	**LS2.C Ecosystem Dynamics, Functioning, and Resilience** — Ecosystems can recover from disturbance through restoration, but recovery depends on the strength of species interactions, the presence of keystone species, and whether the system crosses a self-sustaining threshold.
Crosscutting Concept	**Stability and Change** — The self-sustaining threshold represents a critical transition point — small changes near the threshold produce large shifts in system behavior, demonstrating how stability and change are interconnected in complex systems.

CAST Testing Alignment

MULTIPLE CHOICE:

A restoration team reintroduces beavers to a degraded watershed and monitors recovery for 5 years. Native species diversity is increasing and invasive species are declining, but the self-sustaining threshold has not yet been reached. What should the team recommend? A) Withdraw funding — the ecosystem is improving on its own B) Continue funding — stopping now risks collapse C) Double the number of beavers to speed up the timeline D) Remove all invasive species manually instead of waiting

CONSTRUCTED RESPONSE:

Using your 9-component model, explain why the Early Withdrawal scenario produces different outcomes depending on WHEN funding is stopped. Include the concepts of reinforcing feedback loop and self-sustaining threshold in your answer, and use specific simulation data as evidence.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

This lesson builds on Level 1's keystone species model by expanding from a 4-component removal-and-recovery system to a 9-component restoration system with reinforcing feedback loops, competing pressures, and tipping points. Students now experience the full complexity of real-world ecosystem restoration decisions.

The Science of Ecosystem Restoration

Ecosystem restoration is one of the most important applied sciences of the 21st century. The UN declared 2021-2030 the Decade on Ecosystem Restoration, recognizing that reversing environmental degradation is essential for climate, biodiversity, and human well-being. Restoration ecology combines biology, hydrology, soil science, and data analytics to design recovery plans for damaged landscapes.

Beaver Reintroduction as Restoration Strategy

Across North America and Europe, scientists are reintroducing beavers as a cost-effective restoration strategy. In degraded watersheds, beavers rebuild dams, raise water tables, filter sediment, create wetlands, and dramatically increase biodiversity — all without ongoing human labor. Studies in Washington, Oregon, Utah, and the UK show that beaver reintroduction can restore watersheds at a fraction of the cost of engineered solutions. A single beaver family can restore 5-15 acres of wetland habitat.

Tipping Points and Self-Sustaining Ecosystems

Ecological research shows that recovering ecosystems exhibit tipping points — thresholds beyond which the system switches from requiring active management to maintaining itself. Before the tipping point, removing support causes collapse. After the tipping point, the reinforcing feedback loops between species diversity, habitat quality, and water health are strong enough to sustain the ecosystem independently. Identifying this threshold is one of the hardest problems in restoration ecology.

Why Restoration Projects Fail

The most common cause of restoration failure is premature withdrawal of support. Political cycles (4-year election cycles) are shorter than ecological recovery timelines (10-30 years). When funding is cut before the self-sustaining threshold is reached, ecosystems collapse back to degraded states, wasting all prior investment. Data-driven monitoring allows scientists to identify exactly where a system sits relative to the threshold, providing evidence-based arguments for continued support.

Invasive Species and Recovery

Invasive species are one of the biggest obstacles to restoration. In degraded ecosystems, invasive plants and animals exploit the open niches left by native species decline. As restoration progresses and native species reclaim their niches, invasive pressure naturally decreases — but only if native recovery is strong enough. If restoration is underfunded, invasive species can lock the ecosystem into a degraded state permanently.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L

Locate the System

Students identify Restoration Investment, Keystone Species Reintroduced, and Human Development Pressure as external components — conditions set by human decisions. Habitat Quality, Native Species Diversity, Invasive Species Pressure, Water Quality, Recovery Rate, and Self-Sustaining Threshold are internal components that change based on the interactions between externals and each other.

E

Establish Relationships

Students trace the key relationships: Restoration Investment positively affects Habitat Quality. Keystone Species Reintroduced positively affects both Habitat Quality and Water Quality. Habitat Quality positively affects Native Species Diversity (and Native Species Diversity feeds BACK into Habitat Quality — reinforcing loop). Habitat Quality negatively affects Invasive Species Pressure. Human Development Pressure negatively affects Habitat Quality. All internal components feed into Recovery Rate and Self-Sustaining Threshold.

V

Visualize & Model

Students build the full 9-component model with three external inputs, six internal components, a reinforcing feedback loop, and a threshold output that aggregates system health.

E

Evaluate Outcomes

Students run four scenarios — Full Investment, Underfunded, Development Pressure, and Early Withdrawal — to test how different restoration strategies and pressures affect whether the system crosses the self-sustaining threshold.

R

Revise & Extend

Students add Climate Variability, Community Engagement, or Soil Microbiome Health to explore how additional real-world factors complicate restoration timelines and outcomes.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"In Level 1, we watched an ecosystem COLLAPSE when we removed the keystone species. Today we reverse the question: how do we put it back TOGETHER — and how do we know when it can stand on its own?"*

Watch For

Show before-and-after satellite imagery of a real beaver reintroduction site (e.g., Bridge Creek, Oregon). Ask: What differences do you see?

Slide 2: Learning Objectives

3 min

Say: *"Nine components today — the biggest model in our series. We are adding investment, invasive species, water quality, and the concept of a self-sustaining threshold."*

Watch For

Pre-teach 'tipping point' and 'self-sustaining threshold' with a physical analogy: push a ball up a hill — if you stop pushing before the top, it rolls back. If you push it past the top, it rolls forward on its own.

Slide 3: Big Question

3 min

Say: *"Imagine you are in charge of a restoration project with a limited budget. When is it safe to stop spending — and what happens if you guess wrong?"*

Watch For

Think-pair-share: Students discuss what they think 'self-sustaining' means. How would you measure it? What data would convince you?

Slide 4: LEVER Framework

3 min

Say: *"This model has two new features: a reinforcing feedback LOOP and a THRESHOLD — a tipping point where the system shifts from needing help to helping itself."*

Watch For

Draw the reinforcing loop (Habitat Quality and Native Diversity) on the board. Ask: What happens when this loop is spinning fast? What if it stalls?

Slide 5: Activity 1: Components

8 min

Say: *"Three things humans DECIDE, six things the ecosystem PRODUCES. But watch out — one of those externals is working AGAINST recovery."*

Watch For

Guide component sorting. Key discussion: Human Development Pressure is external but NEGATIVE. Unlike the other two externals, more of this is BAD. Why did we include it?

Slide 6: Activity 2: Connections

10 min

Say: *"This is where it gets real. Restoration and Keystone Reintroduction push Habitat Quality UP. Human Development pushes it DOWN. Habitat Quality and Native Diversity reinforce each other. And everything flows into the Self-Sustaining Threshold."*

Watch For

Build the relationship web step by step. Key moment: draw the reinforcing loop between Habitat Quality and Native Diversity. Ask: What kind of loop is this? What happens when it activates?

Slide 7: Activity 3: Simulation

12 min

Say: *"Four strategies, four outcomes. Full investment, underfunded, fighting development pressure, and the critical one — what happens when we stop too early?"*

Watch For

Run all four scenarios. Require written predictions BEFORE each run. The Early Withdrawal scenario is the climax — run it last. Watch the Self-Sustaining Threshold carefully.

Slide 8: Discoveries

5 min

Say: *"The tipping point is real. Cross it and the ecosystem takes over. Fall short and everything you invested is lost. This is why data-driven monitoring is not optional — it is essential."*

Watch For

Summarize findings. Discuss: How does this change how you think about funding timelines? Why is the Early Withdrawal scenario the most important lesson for policymakers?

Slide 9: STEM Challenge

12 min

Say: *"Now you are the restoration ecologists. Pick a real degraded ecosystem in our state, design a recovery plan, and pitch it to our conservation board."*

Watch For

Form teams, assign research roles, distribute pitch guidelines. Teams research a real degraded site, apply model logic to design a plan with budget and timeline, and prepare a 5-minute pitch.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Restoration Investment, Keystone Species Reintroduced, Human Development Pressure
- **INTERNAL:** Habitat Quality, Native Species Diversity, Invasive Species Pressure, Water Quality, Recovery Rate, Self-Sustaining Threshold

Relationships

- **Restoration Investment to Habitat Quality:** POSITIVE (+) — Higher investment funds planting native vegetation, removing invasive species, engineering water features, and monitoring progress — all of which directly improve habitat structure and function.
- **Keystone Species Reintroduced to Habitat Quality:** POSITIVE (+) — Reintroduced keystone species like beavers physically reshape the landscape — building dams, creating ponds, raising water tables, and generating complex habitat structure that no amount of human engineering can replicate as efficiently.
- **Keystone Species Reintroduced to Water Quality:** POSITIVE (+) — Beaver dams slow water flow, allowing sediment to settle, filtering pollutants, increasing dissolved oxygen through aeration, and raising the water table — all measurable improvements in water quality.
- **Human Development Pressure to Habitat Quality:** NEGATIVE (-) — Construction, agriculture, roads, and pollution degrade habitat by fragmenting landscapes, introducing pollutants, reducing water availability, and disturbing wildlife. This force works AGAINST restoration investment.
- **Habitat Quality to Native Species Diversity:** POSITIVE (+) — Higher habitat quality provides more niches, better food sources, more nesting sites, and cleaner water — all of which support a greater variety of native species colonizing the recovering ecosystem.
- **Native Species Diversity to Habitat Quality:** POSITIVE (+) — REINFORCING FEEDBACK LOOP — More native species improve habitat through pollination, seed dispersal, nutrient cycling, soil stabilization, and pest control. Biodiversity itself is a habitat-building force. This creates a reinforcing loop that ACCELERATES recovery once it gets going.
- **Habitat Quality to Invasive Species Pressure:** NEGATIVE (-) — As habitat quality improves and native species fill ecological niches, invasive species face increasing competition. Healthy native ecosystems naturally resist invasion through competitive exclusion.
- **Habitat Quality, Native Species Diversity, Invasive Species Pressure, and Water Quality to Recovery Rate:** COMPOSITE — Recovery Rate is driven by the combined health of all internal biotic and abiotic indicators. High Habitat Quality, high Native Diversity, low Invasive Pressure, and high Water Quality all accelerate recovery.
- **Recovery Rate to Self-Sustaining Threshold:** POSITIVE (+) with THRESHOLD BEHAVIOR — As Recovery Rate accumulates over time, the system approaches the Self-Sustaining Threshold — the tipping point where internal feedback loops are strong enough to maintain the ecosystem without external investment. Below the threshold, the system collapses without support. Above it, the system sustains itself.

Key Discoveries

- Ecosystem restoration is not linear — the reinforcing loop between Habitat Quality and Native Species Diversity creates accelerating recovery once it gets going
- Invasive Species Pressure drops as Habitat Quality rises — healthy native ecosystems naturally resist invasion
- Keystone species reintroduction is a force multiplier — beavers improve Water Quality AND Habitat Quality simultaneously
- Human Development Pressure can overwhelm even well-funded restoration if it is severe enough
- The Self-Sustaining Threshold is a real tipping point — ecosystems that cross it continue recovering even without funding, but ecosystems that do not cross it collapse when funding stops

- Early withdrawal of funding is the most common cause of restoration failure — timing matters as much as money

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Ecosystems recover naturally if you just leave them alone"	Some mildly degraded ecosystems can self-recover, but severely degraded systems often cannot. When keystone species are gone, invasive species dominate, and soil and water systems are damaged, the ecosystem is trapped in a degraded state. Active restoration — reintroducing species, removing invasives, engineering water features — is required to push the system past the tipping point where natural recovery processes can take over.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"More money always means better restoration"	The Underfunded scenario shows that inadequate funding fails, but the Development Pressure scenario shows that even MAXIMUM funding can fail if opposing forces are strong enough. Money is necessary but not sufficient — the relationship between investment and development pressure matters more than the raw dollar amount. Smart allocation beats bigger budgets.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Once you see improvement, the ecosystem is saved"	Early improvement does NOT mean the self-sustaining threshold has been reached. The Early Withdrawal scenario demonstrates this clearly: the ecosystem can look like it is recovering — rising habitat quality, increasing species diversity — but if the reinforcing feedback loop is not yet strong enough to sustain itself, cutting support causes collapse. Improvement is a sign of PROGRESS, not completion. Only data showing that internal feedback loops are self-maintaining indicates true recovery.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Invasive species must be completely eliminated for restoration to succeed"	Complete elimination of invasive species is rarely possible or necessary. What matters is reducing invasive pressure enough that native species can outcompete them. As Habitat Quality improves and Native Diversity increases, the ecosystem naturally resists invasion through competitive exclusion. The goal is tipping the balance, not achieving zero invasives.